ANALYSIS REPORT

GUIDE

(Guided User Itinerary & Destination Explorer)

A Personalized Tour Recommendation and Guidance System

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Contents

1 INTRODUCTION **2**

2 CURRENT SYSTEM **2**

3 PROPOSED SYSTEM **2**

3.1 Overview . . . 2

3.1.1 System Architecture . . . 3

3.1.2 System Limitations and Constraints . . . 3

3.1.3 Key Features and Capabilities . . . 3

3.2 Functional Requirements . . . 4

3.2.1 Input and Preference Management . . . 4

3.2.2 Core Route Generation . . . 4

3.2.3 Route Visualization and User Interaction . . . 5

3.2.4 Dynamic Route Modification and Feedback . . . 5

3.2.5 Multilingual Content Delivery and Transparency . . . 5

3.3 Non-Functional Requirements . . . 5

3.3.1 Performance and Scalability . . . 6

3.3.2 Usability and Interface Quality . . . 6

3.3.3 Reliability and Error Handling . . . 6

3.4 System Models . . . 6

3.4.1 Scenarios . . . 7

3.4.2 Use Case Model . . . 8

3.4.3 Object and Class Model . . . 11

3.4.4 Dynamic Models . . . 13

3.4.5 User Interface: Navigational Paths and Screen Mock-ups . . . 19

4 Technologies **23**

4.1 Backend Technologies . . . 23

4.2 Frontend Technologies . . . 24

4.3 External Services . . . 24

5 GLOSSARY **24**

REFERENCES **25**

1 INTRODUCTION

GUIDE (Guided User Itinerary & Destination Explorer) is a context-aware travel planning and guidance platform designed to support personalized itinerary generation for cultural and historical travel within Turkey. The system addresses the fragmentation of existing travel planning tools, where users must rely on multiple disconnected applications for navigation, research, and contextual information.

GUIDE integrates route optimization, itinerary generation, and multilingual content delivery into a unified platform. The system utilizes OpenStreetMap (OSM) [1] for geospatial data, Open Source Routing Machine (OSRM) [2] for route calculation, Wikimedia services [3] for descriptive content, Google Places [4] for visual media, and Google Text-to-Speech [5] for multilingual audio generation. By considering user preferences such as trip duration, daily distance constraints, and POI categories, GUIDE generates optimized multi-day itineraries supported by interactive map visualization and audio guidance in Turkish, English, and German.

2 CURRENT SYSTEM

As of the final stage of the project (April 2026 - updated), the GUIDE system has been fully implemented and is operational as a web-based application. The system integrates backend and frontend components to provide end-to-end functionality, including preference-aware route generation, interactive itinerary visualization, and multilingual content delivery.

The backend system implements the core logic for POI filtering, candidate route generation, scoring, and itinerary construction. It is supported by a relational database and external services for routing, content acquisition, and multimedia processing. The frontend provides an interactive user interface that enables users to input preferences, visualize generated routes on a map, and dynamically modify itineraries through actions such as adding, removing, or reordering stops.

The system has been tested under typical usage scenarios, and core functionalities are verified to operate within the defined performance constraints. Route generation is completed within acceptable response times, and the system provides a consistent user experience across its main features.

Although the system is not deployed as a publicly accessible production service, it is fully functional within the development environment and ready for demonstration and evaluation.

3 PROPOSED SYSTEM

3.1 Overview

GUIDE (Guided User Itinerary & Destination Explorer) is proposed as an integrated, context-aware travel planning platform that addresses the fundamental limitations of existing fragmented solutions. The system combines intelligent route generation, preference-aware route recommendations, interactive map visualization, and multilingual audio guidance within a unified web-based interface. The core innovation lies in integrating multiple technologies and data sources to provide comprehensive trip planning support. Unlike existing tools, GUIDE is

designed to minimize planning effort while maximizing trip quality.

3.1.1 System Architecture

GUIDE is designed using a layered architectural approach that aims to keep system responsibilities clearly separated while supporting scalability and maintainability. The architecture consists of four main layers: presentation, application, integration, and data. The presentation layer is responsible for user interaction and map-based visualization through a responsive web interface. The application layer implements the core system logic, including route generation, POI filtering, and itinerary construction. External services such as routing and text-to-speech are handled within the integration layer to isolate third-party dependencies. The data layer manages structured POI metadata and associated multimedia content. A more detailed discussion of the architectural design, including component-level diagrams and design decisions, is provided in the High-Level Design (HLD) report [6].

3.1.2 System Limitations and Constraints

The design and implementation of GUIDE are influenced by a combination of technical, infrastructural, functional, and academic constraints. These include performance limits on route generation, a restricted geographic scope limited to Turkiye, and the absence of real-time traffic, road condition, and dynamic POI operational data. The system relies on static maps and content sources, with POI information curated during development and not continuously updated. Infrastructure constraints affect storage capacity, external map tile dependencies, and deployment resources, while functional scope limitations define supported languages, data sources, and session-based user interaction without persistent accounts. In addition, academic constraints such as project timeline, documentation standards, team capacity, and a preference for open-source technologies shape design decisions and implementation strategies. A comprehensive and detailed specification of these constraints, including their rationale and implications, is provided in the Project Specification Report document [7].

3.1.3 Key Features and Capabilities

Heuristic-Based Route Generation: GUIDE generates personalized multi-day itineraries by creating candidate routes from filtered POI pools based on user-defined categories, geographic constraints, and trip preferences. Candidate routes are evaluated using heuristic criteria such as estimated travel distance, visit durations, and route composition, while final route distances are computed using the Open Source Routing Machine (OSRM) with OpenStreetMap data [8], [9].
Interactive Map Visualization: Generated itineraries are presented through an interactive map interface that displays day-based route segments, POI markers, and connecting paths. Users can explore routes through standard map interactions such as zooming, panning, and marker selection, enabled by the Leaflet.js mapping library [10].
Multilingual Content Delivery: The system provides textual and audio POI information in Turkish, English, and German. Audio guidance is pre-generated using a text-to-speech service to improve accessibility and support users with different language preferences [11].
Dynamic Route Modification: Users may adjust suggested itineraries by removing, adding, or reordering POIs. The system automatically recalculates optimized routes based on the updated

POI set while preserving overall itinerary consistency.

3.2 Functional Requirements

Functional requirements define the core system behaviors that enable GUIDE to generate personalized itineraries and provide destination guidance. Rather than listing individual requirements, this analysis groups them. The groups are presented in Table 1. The complete and formal specification of all functional requirements is provided in the Project Specification Report[7].

Table 1: Functional Requirement Groups and Scope

Requirement Group	**Covered High-Level Requirements**	**Primary Focus**
Input and Preference Management	FR-H01	User input collection, validation, and preference specification
Core Route Generation	FR-H02, FR-H07	Route generation, optimization, itinerary construction, scoring, and visit duration estimation
Route Visualization	FR-H04, FR-H06	Map-based route display and user interaction
Route Modification and Feedback	FR-H03, FR-H08	User-controlled route adjustments and error handling
Multilingual Content and Transparency	FR-H05, FR-H09	Accessibility, text-to-speech support, and data source disclosure

3.2.1 Input and Preference Management

This requirement group covers the mechanisms through which users provide the necessary inputs for route generation, including destination selection, trip duration, distance constraints, category filters, and exclusion preferences. These requirements correspond primarily to FR-H01 and its related low-level requirements. Input management forms the entry point of the system and directly influences all subsequent processing stages.

Dependencies: All route generation, itinerary optimization, and visualization requirements depend on the validity and completeness of user inputs. Invalid or inconsistent inputs prevent further system operation.

Analysis Considerations: While input handling is not computationally complex, robust validation and clear error feedback are essential to prevent infeasible routes and ensure a smooth user experience. This group also establishes a strong dependency between functional and usability-related requirements.

3.2.2 Core Route Generation

This group represents the central functionality of GUIDE and includes requirements related to preference-aware route generation and day-by-day itinerary organization. These behaviors correspond to FR-H02, FR-H07, and their associated low-level requirements.

Dependencies: Core route generation depends on validated user inputs, availability of POI metadata, and reliable routing and map services. It also interacts closely with performance requirements due to the computational cost of candidate route generation and evaluation.

Analysis Considerations: This group introduces the highest technical complexity in the system. Generating high-quality routes while satisfying user preferences and meeting performance targets requires careful algorithm design. Candidate itineraries are constructed and evaluated using a heuristic approach, where multiple alternatives are scored and ranked based on factors such as POI popularity, distance constraint compliance, category diversity, and itinerary size, and the highest-scoring itinerary is selected. The use of heuristic distance estimation and city-specific route distance constraints is a direct consequence of these challenges.

3.2.3 Route Visualization and User Interaction

This requirement group addresses how generated routes are presented to the user and how users interact with them. It includes requirements for map-based route visualization, user interaction with map elements, and destination information display, corresponding to FR-H04 and FR-H06.

Dependencies: Visualization requirements depend on successful route generation and the availability of geospatial data. They also rely on external mapping libraries and frontend responsiveness.

Analysis Considerations: Although visualization does not alter route logic, it has a significant impact on perceived system quality. Clear visual structure and intuitive interaction are critical for user comprehension and trust in generated itineraries.

3.2.4 Dynamic Route Modification and Feedback

This group includes requirements that allow users to modify suggested routes and receive meaningful system feedback. It primarily corresponds to FR-H03 and FR-H08.

Dependencies: Route modification depends on the availability of a previously generated route and requires seamless reintegration with the route optimization logic.

Analysis Considerations: Supporting user intervention improves transparency and user autonomy but increases system complexity. Allowing users to remove, add, and reorder POIs requires the system to dynamically recompute routes while preserving consistency and feasibility. Error handling and feedback mechanisms must be carefully designed to maintain system stability during replanning operations.

3.2.5 Multilingual Content Delivery and Transparency

This group covers requirements related to text-to-speech functionality and transparency of data sources, corresponding to FR-H05 and FR-H09.

Dependencies: These requirements depend on the availability of curated POI content and pre-generated audio assets.

Analysis Considerations: Multilingual support enhances accessibility but introduces additional content management and resource constraints. Explicit data source disclosure improves user trust and aligns with ethical and academic transparency standards.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes that constrain how GUIDE's functional requirements are implemented. These requirements influence architectural decisions rather than specifying direct system behaviors. The focus and group of the requirements are

presented in Table **??**.

Table 2: Non-Functional Requirement Groups and Scope

Requirement Group	**Covered High-Level Requirements**	**Primary Focus**
Performance Constraints	PR-H01	Route generation latency and response time limits to maintain user engagement
Usability and Interface Quality	EIR-H01	Web-based interface usability, responsive design, and intuitive user interaction
Reliability and Error Handling	FR-H08*	System stability, fault tolerance, graceful error handling, and meaningful user feedback
Transparency and Trust	FR-H09*	Clear disclosure of data sources and system behavior to users

**Although FR-H08 and FR-H09 are specified as functional requirements in the Project Specification Report, they are analyzed here due to their impact on non-functional quality attributes such as reliability, usability, and user trust.*

3.3.1 Performance and Scalability

Performance requirements focus on route generation latency , corresponding to PR-H01 . These constraints directly affect algorithm selection, system architecture, and deployment strategies.

Analysis Considerations: Maintaining acceptable response times requires efficient algorithms and localized routing services. Scalability considerations motivate stateless backend design and limit system scope to ensure predictable performance under load.

3.3.2 Usability and Interface Quality

Usability-related requirements are reflected primarily in the external interface requirements (EIR-H01 and related low-level requirements). These requirements shape the design of the web-based interface, interaction patterns, and accessibility across devices.

Analysis Considerations: Clear navigation, responsive layout, and intuitive interaction are essential to reduce user effort and support non-technical users. Usability constraints influence frontend technology choices and validation strategies.

3.3.3 Reliability and Error Handling

Reliability requirements ensure consistent system behavior in the presence of invalid inputs, network issues, or partial system failures. These concerns are closely linked to functional error handling requirements.

Analysis Considerations: Graceful degradation and informative error feedback are critical to maintaining user trust and preventing system crashes, especially during route generation and replanning phases.

3.4 System Models

This section presents the system models used to describe the structural and behavioral aspects of the GUIDE system. The models collectively illustrate how the system is used, how its components are organized, and how they interact over time. Scenarios provide representative usage

examples, the use case model captures user–system interactions, object and class models describe the static structure of the system, dynamic models illustrate runtime behavior and message flows, and user interface models present navigational paths and screen-level interactions.

3.4.1 Scenarios

The following scenarios present representative usage examples of the GUIDE system. These scenarios illustrate typical system behavior under different user profiles, preferences, and constraints. Collectively, they reflect the full functional scope of the system, including input specification, route generation, itinerary modification, map-based visualization, and multilingual content delivery.

Scenario 1: Erasmus Student Visiting Istanbul
User Type: International student / short-term visitor
Scenario Description: Kayrahan is an Erasmus student staying in Turkiye and plans to spend three days in Istanbul. Upon accessing GUIDE, the user specifies the trip duration (three days), destination (Istanbul), and a preferred daily travel distance in the range of 30–50 km. The user further selects relevant main categories and subcategories according to personal interests, such as the Historical and Museums main categories and selected subcategories including Religious and Ancient.
Based on the provided inputs, GUIDE constructs and evaluates candidate routes using the selected categories, destination constraints, and travel preferences, and generates a three-day itinerary organized on a day-by-day basis. For each destination, GUIDE provides descriptive text and visual content. During the trip, the user selects English audio guidance for POI descriptions. GUIDE also allows the user to remove unwanted destinations and regenerate the route accordingly.
Objective: To generate a culturally rich and time-efficient itinerary that satisfies the user's category preferences and daily travel distance expectations for an international short-term visitor.

Scenario 2: Family-Oriented Domestic Trip Planning
User Type: Domestic user / family trip planner
Scenario Description: Erdem plans a five-day family vacation in Antalya. Using GUIDE, the user enters the destination, trip duration, daily route distance interval, and main and related subcategories, explicitly excluding the Museums main category while selecting the full set of subcategories under the Nature main category.
Based on these inputs, GUIDE informs the user that the selected constraints and category preferences can support a maximum itinerary duration of four days. Since Erdem does not wish to modify the selected category preferences, the user accepts the four-day itinerary option and proceeds accordingly. GUIDE then constructs and evaluates candidate routes and generates a four-day itinerary organized into distinct daily routes. Each destination includes descriptive information, visual content, and Turkish audio narration. GUIDE also provides estimated visit durations for each POI to support flexible daily planning without enforcing strict schedules.
Objective: To support family users in planning practical, preference-aware daily itineraries that emphasize natural and leisure-oriented destinations while maintaining route quality and

satisfying user preferences as completely as possible.

Scenario 3: Language-Oriented Cultural Exploration
User Type: International tourist / language-learning user
Scenario Description: Tuna is a traveler learning German who wishes to explore cultural destinations in Izmir over a two-day period. The user specifies the destination, trip duration, and preferred daily travel distance range. The user also selects relevant main categories and subcategories related to cultural exploration. Based on these inputs, GUIDE constructs and evaluates candidate routes and generates a two-day itinerary organized on a day-by-day basis. For each destination, the system provides textual descriptions, images, and German text-to-speech audio, allowing the user to combine cultural exploration with language practice.
Objective: To enhance cultural travel experiences by integrating multilingual audio guidance that supports language learning.

Scenario 4: Local User Exploring Nearby Attractions
User Type: Local resident
Scenario Description: Ebrar is a local resident in Ankara with one available day for exploration. After entering the destination, trip duration, and preferred daily travel distance range, the user selects relevant main categories and subcategories related to cultural and historical exploration. Based on these inputs, GUIDE constructs and evaluates candidate routes and presents a suggested daily itinerary. The user may exclude previously visited destinations, prompting GUIDE to regenerate an updated route. The finalized itinerary is displayed on an interactive map and supported with descriptive content and Turkish audio guidance for each POI.
Objective: To enable local users to discover cultural and historical landmarks within their own cities through preference-aware daily route generation.

3.4.2 Use Case Model

GUIDE's use case model describes the primary interactions between users and the system, structured around the core workflow of preference specification, route planning, route visualization, and destination information access. Each use case is supported by a dedicated diagram to illustrate the corresponding user–system interaction in isolation.

UC-01: Manage Preferences
This use case represents the initial configuration phase of the system. GUIDE provides a structured input interface that allows users to specify essential planning parameters such as destination (city or region), trip duration, maximum travel distance constraints, and category-based POI preferences. The system validates the selected destination to ensure correctness before proceeding. These inputs define the constraints and scope of the itinerary generation process and directly influence all subsequent system behavior. The interaction flow of this use case is illustrated in Figure 1.

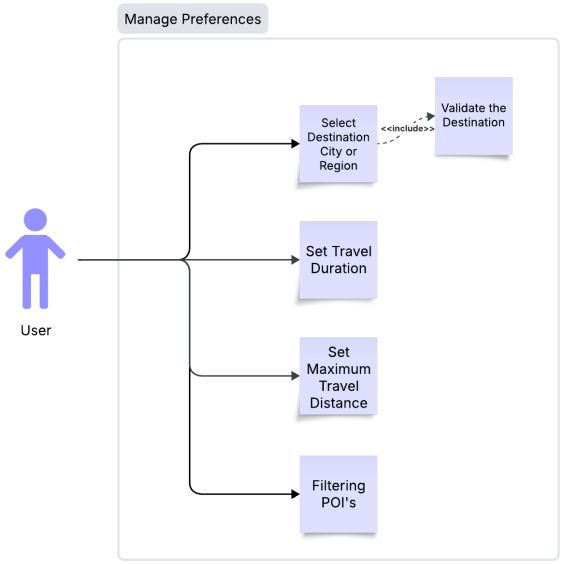


Figure 1: Use Case Diagram for UC-01: Manage Preferences

UC-02: Route Planning

This use case captures the core route planning functionality of the GUIDE system. Based on user-provided inputs and selected filters, the system generates a route by constructing and evaluating candidate itineraries. The resulting itinerary is organized on a day-by-day basis and presented to the user. GUIDE supports iterative refinement by allowing users to add, remove, and reorder POIs within the generated route. Following any modification, the system replans and regenerates the route while maintaining overall itinerary consistency. The interaction flow is shown in Figure 2.

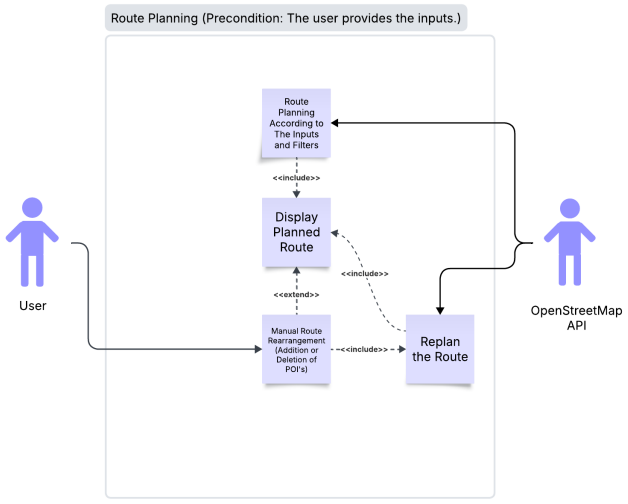


Figure 2: Use Case Diagram for UC-02: Route Planning

UC-03: Route and POI Visualization

This use case focuses on the visualization and exploration of a generated itinerary. The user initiates the process by opening the planned route, after which GUIDE displays the itinerary on

an interactive map with POIs represented as markers and connected through route paths. The map interface enables users to explore the itinerary spatially and interact with individual destinations. Users may click on POI markers to retrieve detailed information about each location. The interaction flow of this use case is illustrated in Figure 3.

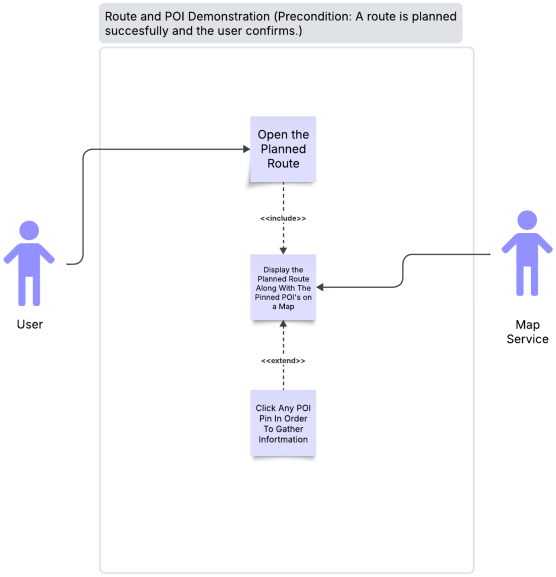


Figure 3: Use Case Diagram for UC-03: Route and POI Visualization

UC-04: POI Information Access

This use case describes how detailed information about individual destinations is accessed after a POI is selected. Upon selecting a POI marker, GUIDE displays a detailed view including textual descriptions, images, and related metadata. The system provides multiple photos and descriptive content for each POI, along with optional multilingual audio guidance in Turkish, English, and German.

All content is retrieved from preprocessed and stored datasets, and no real-time external API calls are performed during user interaction. The interaction flow for this use case is depicted in Figure 4.

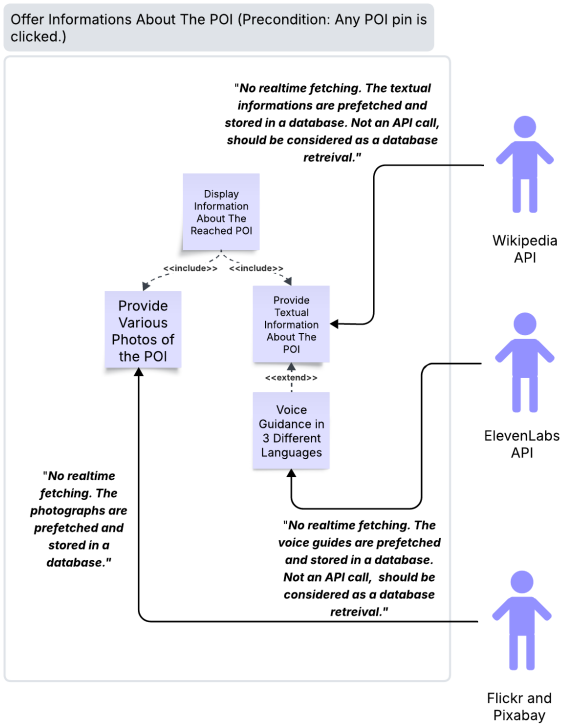


Figure 4: Use Case Diagram for UC-04: POI Information Access

3.4.3 Object and Class Model

This subsection presents the static structure of GUIDE through an object model and a UML class diagram. The object model provides a conceptual view of the main domain entities and how they relate at runtime, while the class diagram refines this view by defining class responsibilities, attributes, and associations.

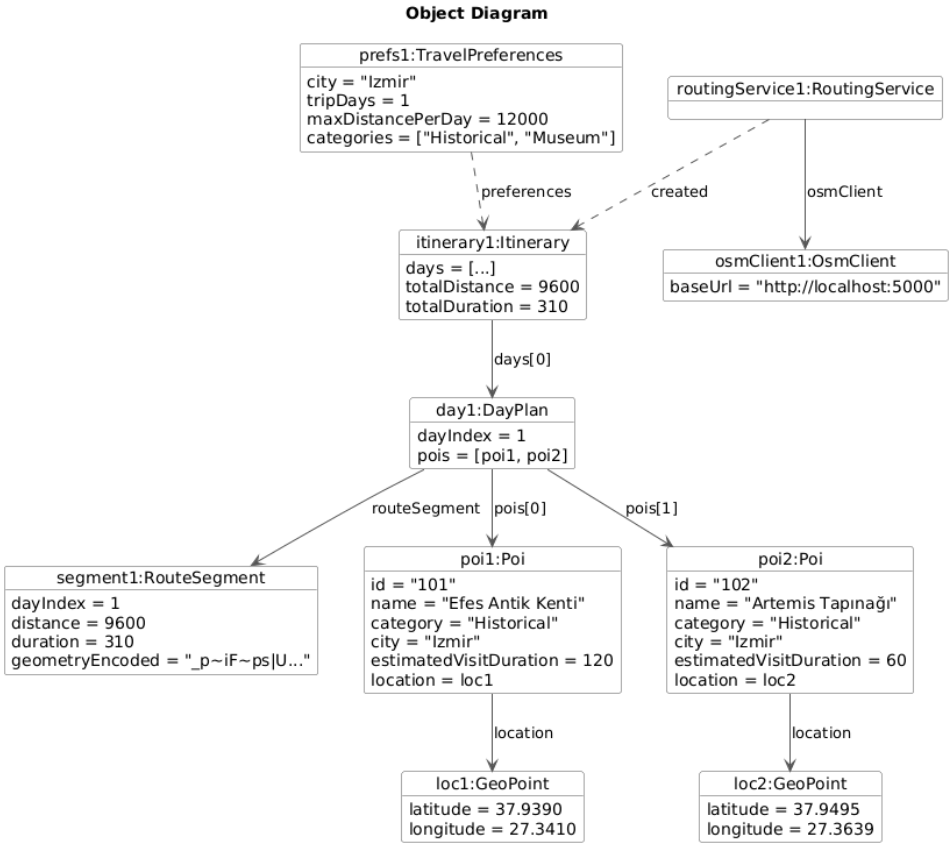


Figure 5: GUIDE Object Model

Figure 5 illustrates the object model of the GUIDE domain, focusing on core entities.

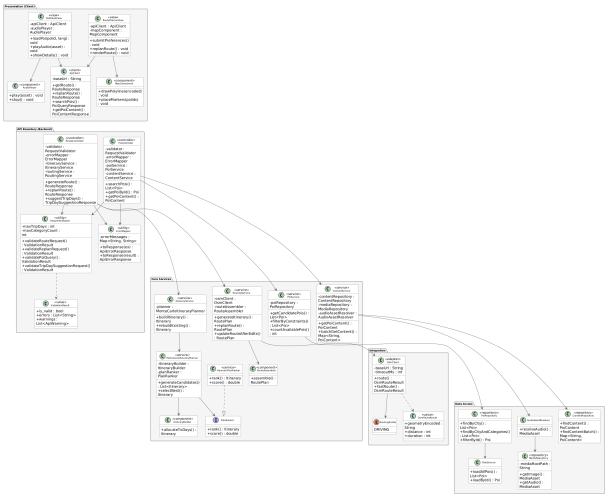


Figure 6: GUIDE System Class Diagram

Figure 6 presents the UML class model of GUIDE, emphasizing the main domain classes and their relationships (e.g., *POI*, *Route*, *RouteDay*, *TravelPreferences*, and supporting data structures such as *Location*). The diagram captures the core data flow from user preferences to itinerary construction and representation.

3.4.4 Dynamic Models

The dynamic behavior of the GUIDE system is described using multiple complementary modeling techniques. Sequence diagrams illustrate message flows and temporal interactions between system components, and activity diagrams model control flow and decision logic across core workflows. These dynamic models focus on representative operational scenarios and do not aim to exhaustively document all possible system interactions; instead, they provide insight into the runtime behavior of GUIDE under typical usage conditions.

Sequence Diagrams:

GUIDE's dynamic behavior is modeled through sequence diagrams illustrating message flows and temporal interactions between system components. The following diagrams represent key operational sequences but do not exhaustively document all possible system interactions.

Sequence 1: Input Collection Phase – Route Planning

This sequence describes the initial interaction phase in which trip planning parameters are collected. GUIDE presents a structured input interface that allows users to specify the destination (city or region), trip duration, maximum travel distance constraint, and category-based POI preferences. Once the inputs are provided, the Web UI sends the collected parameters to the backend service.

The backend evaluates the feasibility of the requested trip duration by querying the database for available POIs based on the selected city and categories. Based on this information, the system computes and returns a suggested feasible number of days.

If the suggested duration does not satisfy the user, the user may adjust category preferences and trigger the evaluation process again. This interaction may repeat iteratively until a satisfactory configuration is reached. Once the user approves the suggested or adjusted duration, the process proceeds to route generation.

The message flow for this interaction is illustrated in Figure 7.

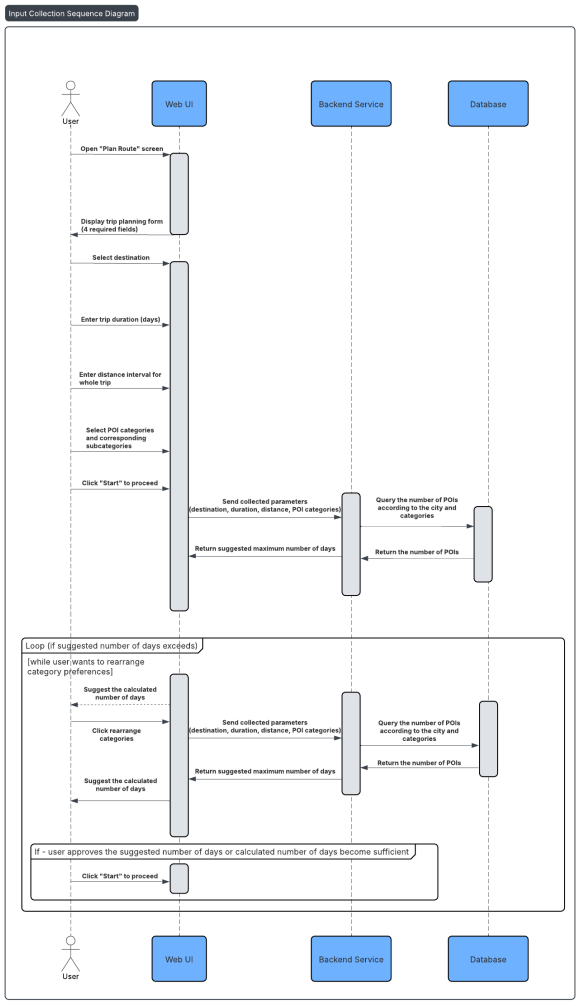


Figure 7: Sequence Diagram – Input Collection Phase

Sequence 2: Route Generation and Replanning

This sequence describes the route generation and replanning process of the GUIDE system. After the user initiates route generation, the Web UI sends the collected parameters to the backend service. The backend retrieves candidate POIs from the database based on the selected filters and preferences. Using these POIs, the backend requests route data from the OSRM service to compute travel paths and distances.

Once the route data is obtained, the backend returns the planned route and associated POIs to the Web UI. The Web UI then renders the route on an interactive map using Leaflet, displaying POI markers and route paths to the user.

If the user modifies the itinerary by adding, removing, or reordering POIs, the updated POI set is sent back to the backend. The backend recomputes the route by requesting updated route data from the OSRM service, and the updated itinerary is returned and rendered again on the map. This process may repeat iteratively until the user is satisfied.

The interaction flow is shown in Figure 8.

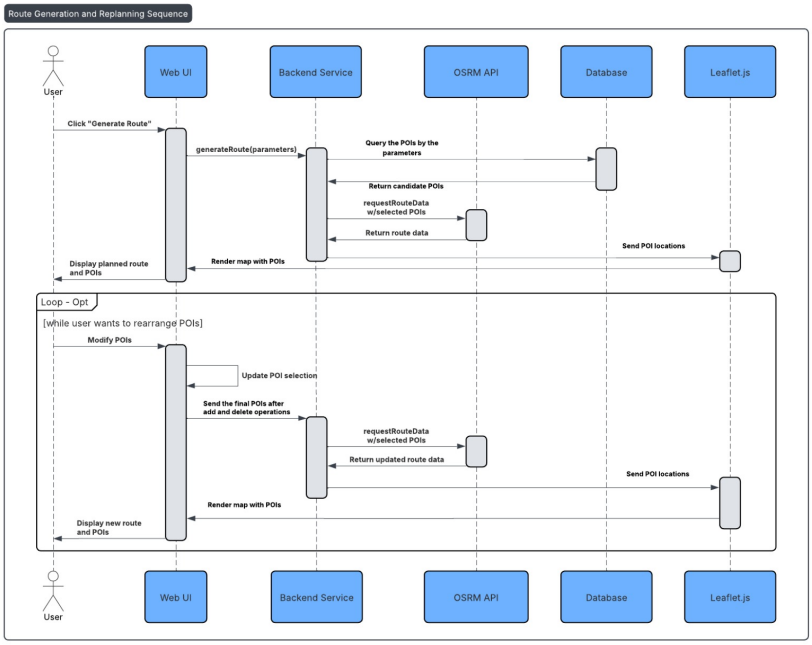


Figure 8: Sequence Diagram – Route Generation and Replanning

Sequence 3: POI Details and Pre-generated Voice Guidance

This sequence describes the retrieval and presentation of detailed POI content. When the user selects a POI marker, the Web UI sends a request to the backend service to retrieve POI details. The backend queries the database for the corresponding textual descriptions, images, and associated metadata, and returns the retrieved data to the Web UI. The POI detail view is then displayed to the user.

Optionally, if the user enables voice guidance and selects a language, the Web UI requests the corresponding audio content from the backend. The backend retrieves the pre-generated audio file reference from the database and returns it to the Web UI, where the audio is played.

All content, including text, images, and audio, is preprocessed and stored in the database, and no real-time external API calls are performed during this interaction. The sequence of interactions is illustrated in Figure 9.

Important Note: POI-related content is planned to be prepared during an offline data preparation phase. Textual descriptions are intended to be obtained via Wikimedia services [12], visual content is planned to be collected using platforms such as Google Places and publicly available image sources [4], and multilingual audio files are planned to be generated using text-to-speech services [13].

All collected content is expected to be preprocessed and stored in the system database prior to runtime. Therefore, the system is designed to avoid real-time external API calls during user interaction, and all content retrieval operations are planned to be performed through internal database access.

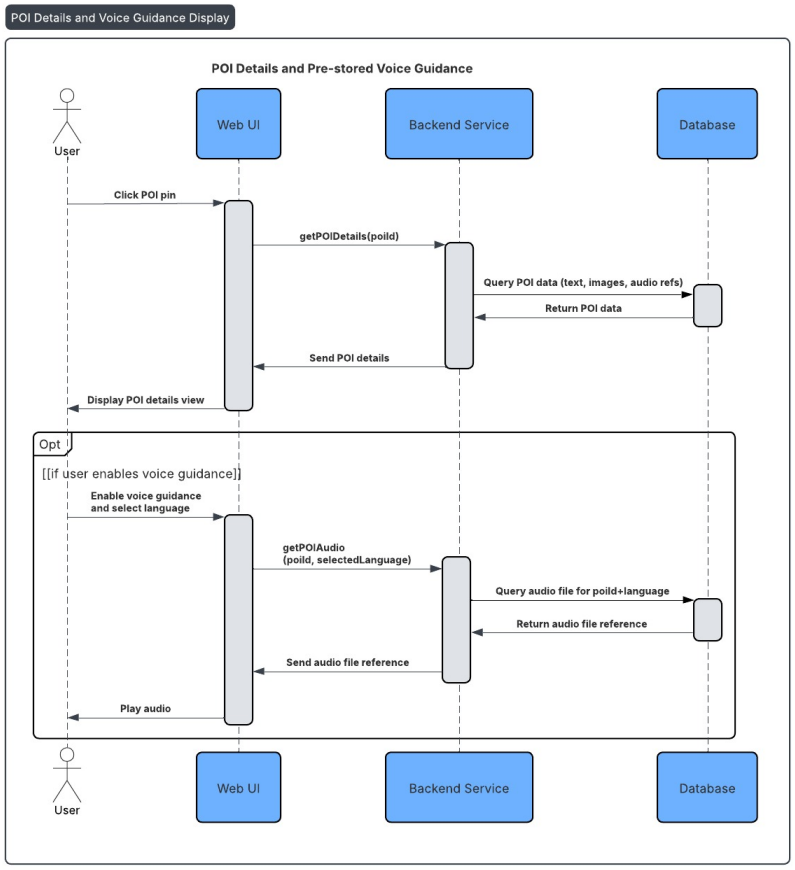


Figure 9: Sequence Diagram – POI Details and Voice Guidance

Activity Diagrams:

Activity diagrams are used to model the control flow and decision logic of GUIDE's core workflows. They illustrate how user actions, system validations, and internal processing steps are coordinated during route planning, POI information retrieval, and route replanning operations.

Activity 1: Complete Route Planning Workflow

Figure 10 represents the end-to-end route planning process, starting from user input collection and preference configuration. The diagram includes decision points related to trip feasibility, such as evaluating whether the selected number of days is sufficient and suggesting adjustments when necessary.

Following input confirmation, the system generates candidate routes, selects an optimal itinerary, and retrieves route details via the OSRM [8]. The resulting itinerary is then organized on a day-by-day basis and presented on the map.

The workflow also models iterative user interaction through two feedback loops: preference adjustment during input configuration and route modification after visualization. Users may add or remove POIs, triggering replanning operations that recompute the route accordingly.

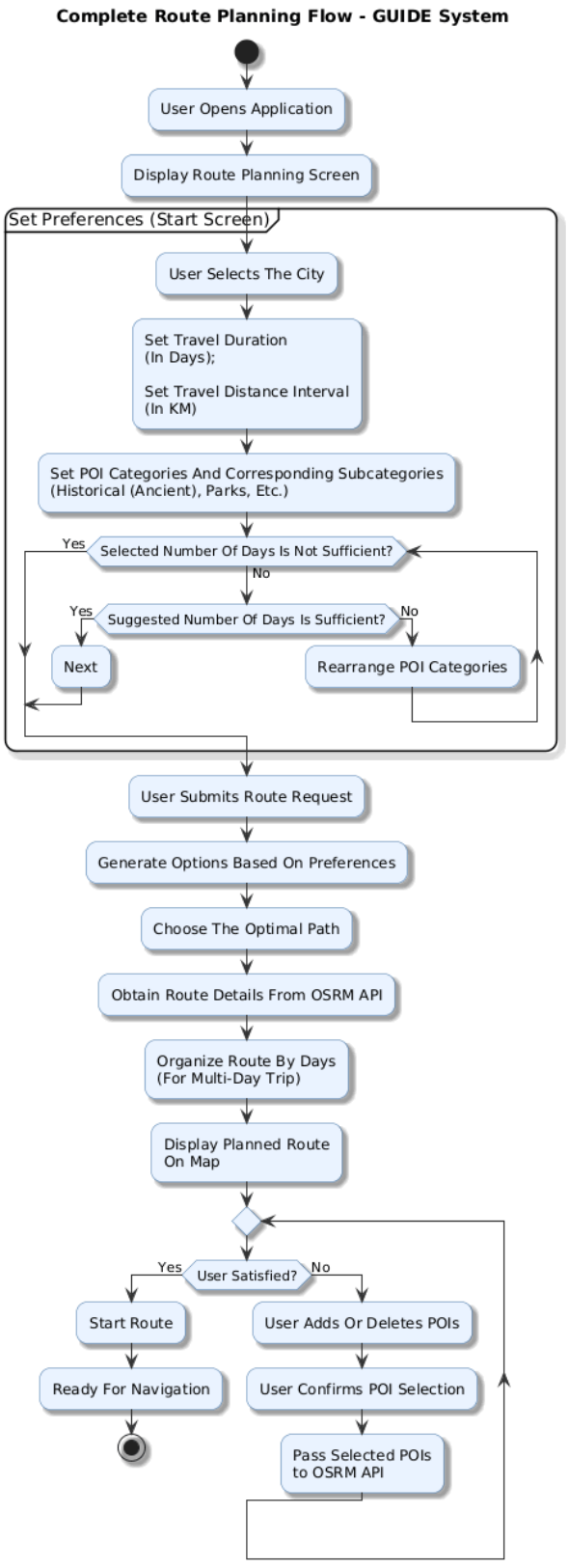


Figure 10: Complete Route Planning Activity Diagram

Activity 2: POI Information Retrieval

Figure 11 illustrates the retrieval and presentation of multimedia content for a selected Point

of Interest. Following POI selection, the system initiates parallel processes to fetch textual descriptions and images from the database, while also checking the availability of pre-generated audio content.

Text and image data are retrieved, formatted, and displayed independently, ensuring efficient response time. In parallel, the system evaluates whether multilingual audio files exist. If audio content is available, the system loads the corresponding language options and allows the user to select a preferred language. Audio playback includes basic user controls (pause, resume, stop) and continues until completion. If audio files are not available, the system falls back to a text-only presentation mode.

After content is displayed, the user may review the information and optionally navigate to another POI or return to the route view. The diagram explicitly models these parallel flows, conditional branches, and user-driven navigation behavior.

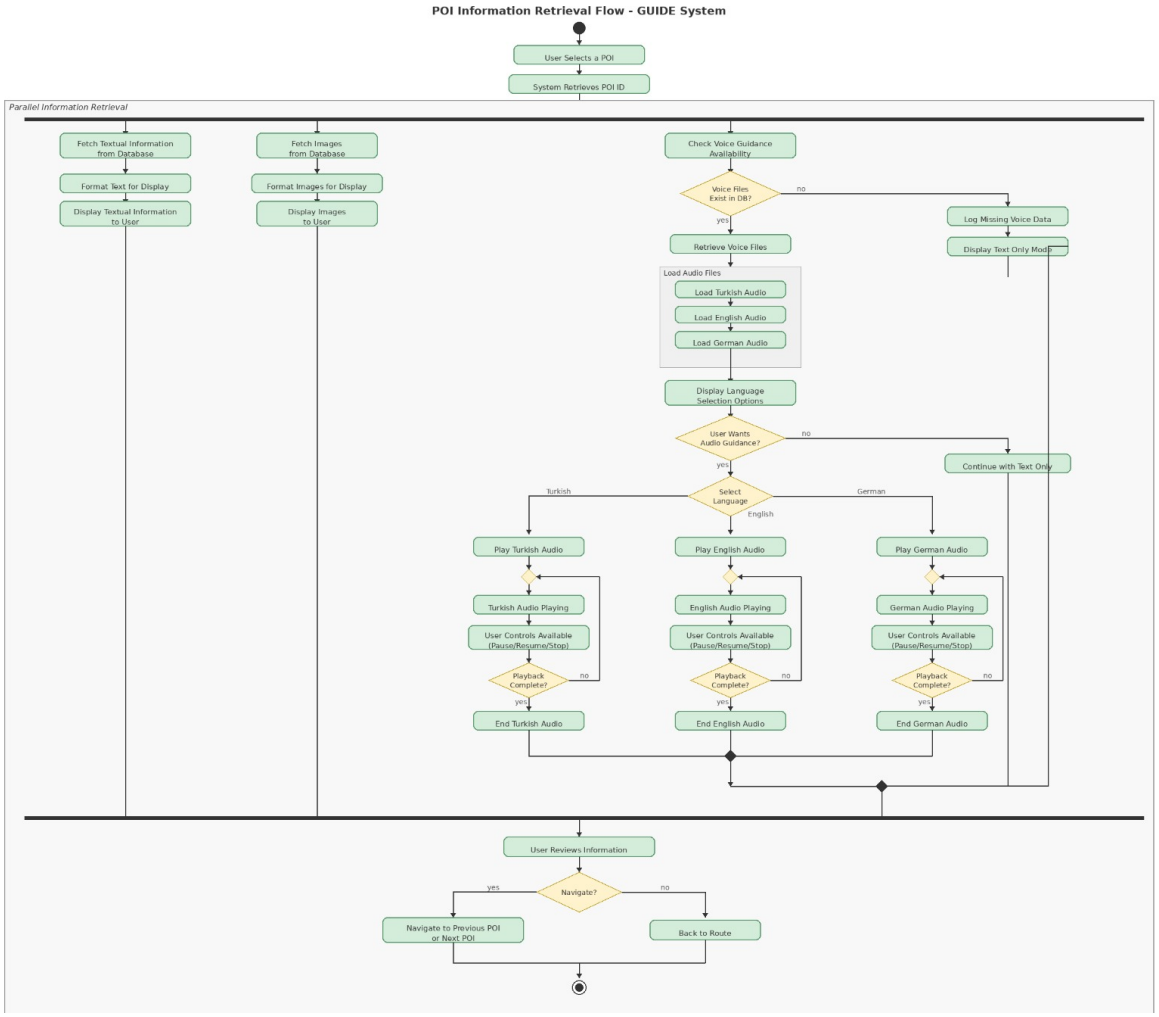


Figure 11: POI Information Retrieval Activity Diagram

3.4.5 User Interface: Navigational Paths and Screen Mock-ups

Figure 12: GUIDE system logo

The GUIDE logo represents the system's core concept of guided exploration and structured travel planning. Its visual identity emphasizes navigation, clarity, and geographic orientation, aligning with the system's objective of generating route-aware itineraries.

Navigational Structure and Screen Overview

GUIDE follows a semi-linear navigation structure that guides users from initial planning to route exploration and iterative refinement. While the general flow progresses forward, users can revisit previous steps (e.g., replanning or returning to the route view), enabling flexible interaction.

Landing Page

The landing page serves as the system entry point and introduces the purpose of GUIDE. It presents the system branding along with concise guidance and two primary navigation options.
Primary Navigation Elements:

- **Start Planning:** Navigates to the trip planning interface
- **How GUIDE Works:** Opens the system overview and workflow explanation page

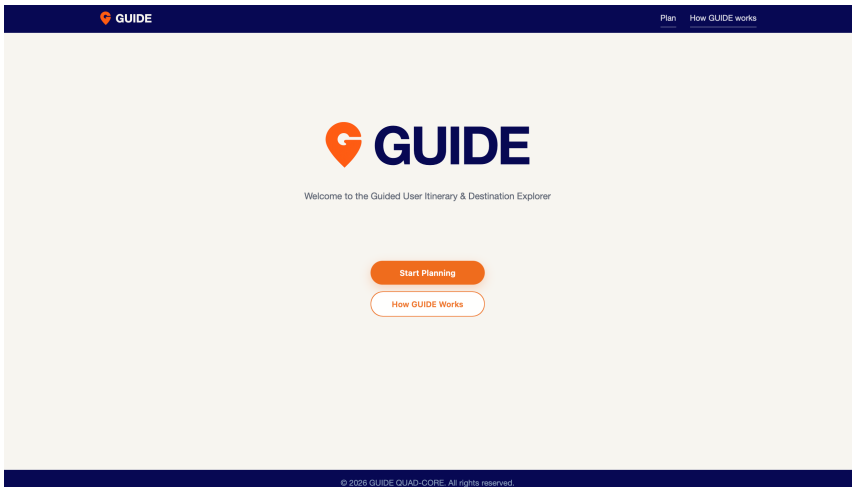


Figure 13: Landing Page

Trip Planning Interface

This screen collects user-defined constraints required for itinerary generation. Users select a city, specify trip duration and daily travel distance, and choose interest categories through an

interactive filtering interface. Available categories are dynamically adapted based on the selected city.

Navigation and Actions:

- **Generate Route:** Submits the input configuration and triggers route generation

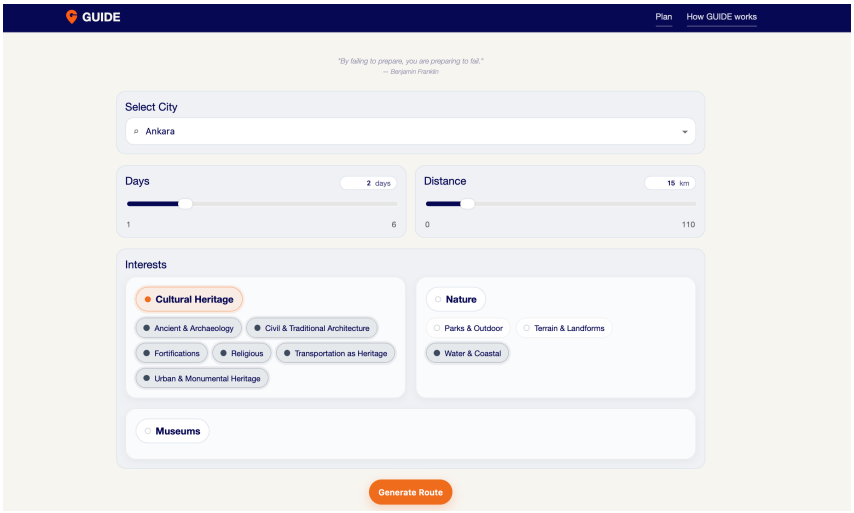


Figure 14: Trip Planning Interface

Route Visualization Page

This page displays the generated itinerary using an interactive map and a structured daily plan. Each day is represented separately, and POIs are visualized as ordered markers connected by computed routes. The interface enables both spatial and sequential exploration of the itinerary.

Navigation and Actions:

- **Select POI (Map or List):** Opens the detailed view of a selected destination
- **Finalize / Continue:** Proceeds with the current itinerary

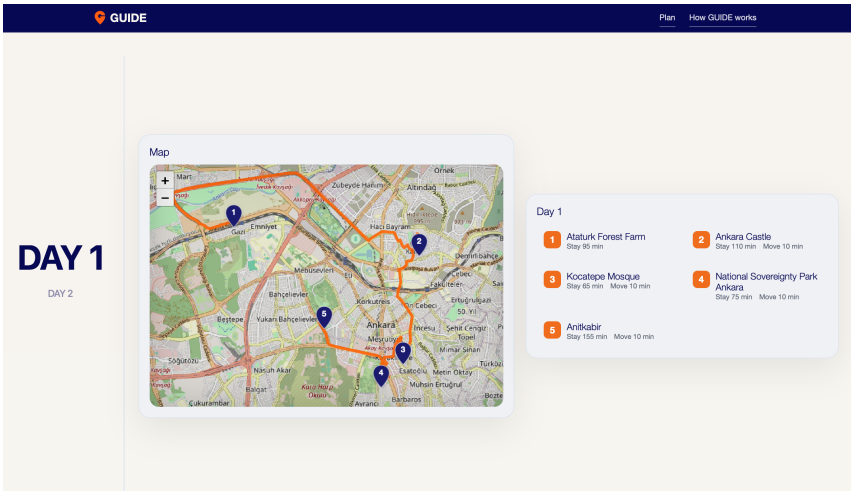


Figure 15: Generated Route Visualization

Route Replanning Interface

The replanning interface enables users to interactively refine the generated itinerary at a per-day level. Each day is represented as a structured plan where points of interest (POIs) can be dynamically modified while maintaining route consistency.

Users can visually inspect the route on the map while simultaneously editing the ordered POI list. Any modification, such as reordering, removing, or adding POIs, triggers a recalculation of the route using the routing engine, ensuring that the updated itinerary remains feasible and optimized.

Key Features:

- **Day-based Editing:** Users can switch between days and edit each day's itinerary independently
- **POI Reordering:** Destinations can be rearranged within a day to customize visit order
- **POI Removal:** Unwanted locations can be removed instantly from the route
- **Search and Add:** Users can search and insert new POIs into a specific day plan
- **Finalize Replan:** Changes are confirmed and the updated route is persisted for further use

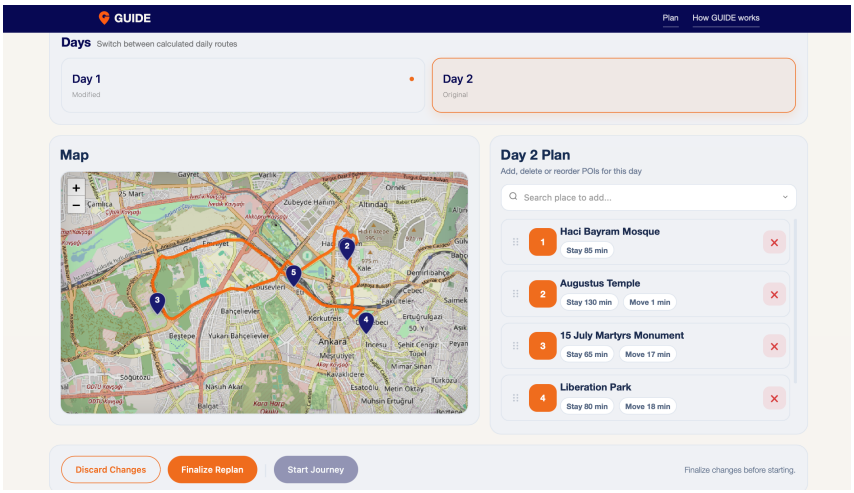


Figure 16: Route Replanning Interface

POI Detail View

The POI detail page provides rich, destination-specific information. It includes images, descriptive text, estimated visit duration, and multilingual audio narration. This view supports immersive exploration while maintaining continuity within the itinerary.

Navigation and Actions:

- **Audio Playback:** Listen to POI descriptions in Turkish, English, or German
- **Back:** Return to the previous screen

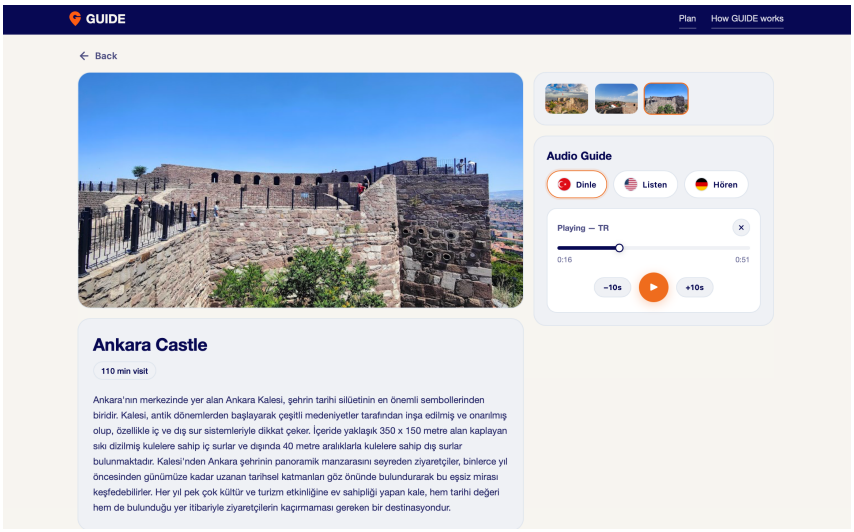


Figure 17: POI Detail View

How GUIDE Works Page

This page presents a high-level explanation of the system's workflow, including data sources, routing logic, and processing steps. It also includes a dataset preview and system flow description to improve transparency and user understanding.

Navigation and Actions:

- **Back to Landing:** Returns to the main page

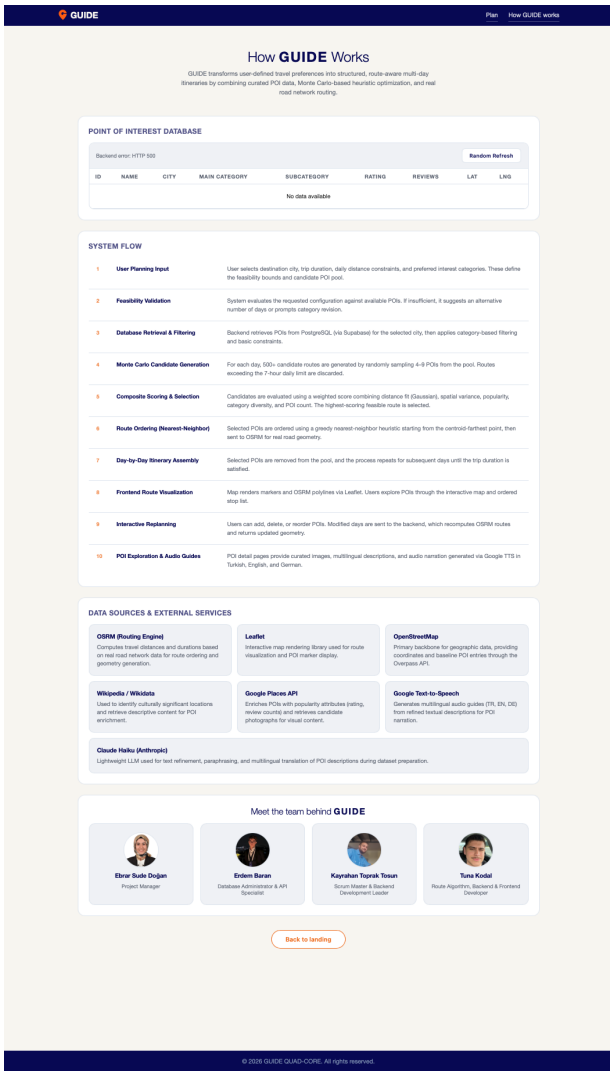


Figure 18: System Workflow and Transparency Page

4 Technologies

This section outlines the technologies and external services planned to be used in the implementation of the GUIDE system. The system is designed as a combination of backend and frontend components, supported by external services for routing, content processing, and multimedia management.

4.1 Backend Technologies

The backend is planned to be implemented using Python due to its flexibility and suitability for algorithmic workflows [14]. Core functionalities such as POI filtering, candidate route generation, scoring, and itinerary construction are expected to be developed as modular services.

A relational database system shall be used for structured data storage, with PostgreSQL as the underlying technology [15]. Supabase is planned to be adopted as a backend-as-a-service solution built on PostgreSQL, providing API generation and simplified data access [16].

Multimedia content, including images and audio files, shall be stored using cloud-based ob-

ject storage services such as Amazon S3, enabling scalable and efficient media management [17].

Version control is planned to be maintained using Git, with the project hosted on GitHub [18], [19]. Development is expected to be carried out using modern integrated development environments such as PyCharm [20].

4.2 Frontend Technologies

The frontend is planned to be implemented using React to enable interactive and component-based user interface development [21]. The application shall be developed using JavaScript, HTML, and CSS, and executed within a Node.js environment during development and build processes [22].

The frontend shall communicate with the backend via RESTful APIs, exchanging data in JSON format [23], [24]. The interface is expected to support dynamic user interactions such as itinerary visualization and route modification.

4.3 External Services

The system is planned to integrate several external services to support routing, content acquisition, and multimedia processing.

For routing, the Open Source Routing Machine (OSRM) shall be used in conjunction with OpenStreetMap data to compute travel distances and durations [1], [25].

For content acquisition, descriptive information shall be retrieved from publicly available sources such as Wikimedia services, and may be supplemented with controlled web scraping where necessary [3].

Visual media is planned to be supported through services such as Google Places [4], while multilingual audio content shall be generated using Google Text-to-Speech services [5].

Content processing and refinement may be supported by lightweight language models during dataset preparation. However, all generated content shall be reviewed and curated offline before being incorporated into the system to ensure consistency and reliability.

Overall, these technologies are selected to enable a modular, scalable, and maintainable system architecture while balancing the use of open-source tools and external service integrations.

5 GLOSSARY

POI (Point of Interest) A destination or location of potential interest to travelers (e.g., museum, historical site, natural landmark).

OSRM (Open Source Routing Machine) Open-source routing engine using OpenStreetMap data to calculate optimal routes.

TTS (Text-to-Speech) Technology converting written text into spoken audio.

OSM (OpenStreetMap) Collaborative open-source geographic database providing map data.

Itinerary Planned sequence of destinations and routes for a trip, organized by days.

Day-Based Segmentation Division of multi-day trips into balanced daily schedules with distance/time constraints.

Polyline Sequence of connected line segments representing a route on a map.

Responsive Design Web design approach adapting layouts to different screen sizes (desktop, tablet, mobile).

Session-Based Architecture System design where user data exists only during an active session, not persistently stored.

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Note: Spell checking and basic grammatical verification were performed using Grammarly: (https://www.grammarly.com).

END OF ANALYSIS REPORT

GUIDE (Guided User Itinerary & Destination Explorer)
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